PYGEOVISION · GEOSPATIAL ANALYSIS REPORT

Building Footprints

n/a (published dataset) · 2.8 km²

604	13.8 ha	5.0 %	219.1
Buildings	Total footprint area	Share of the area covered by buildings	Buildings per km²

Executive summary

- 604 buildings from Microsoft ML Building Footprints fall inside the area, covering 13.8 ha (5.0 % of the area; 219 per km²).
- These are the dataset's polygons, not detections on your own imagery.

> Microsoft Global ML Building Footprints: machine-learning detections on Bing/Maxar/Vexcel imagery of various dates (about 1% false positives reported by the project); building heights where available.

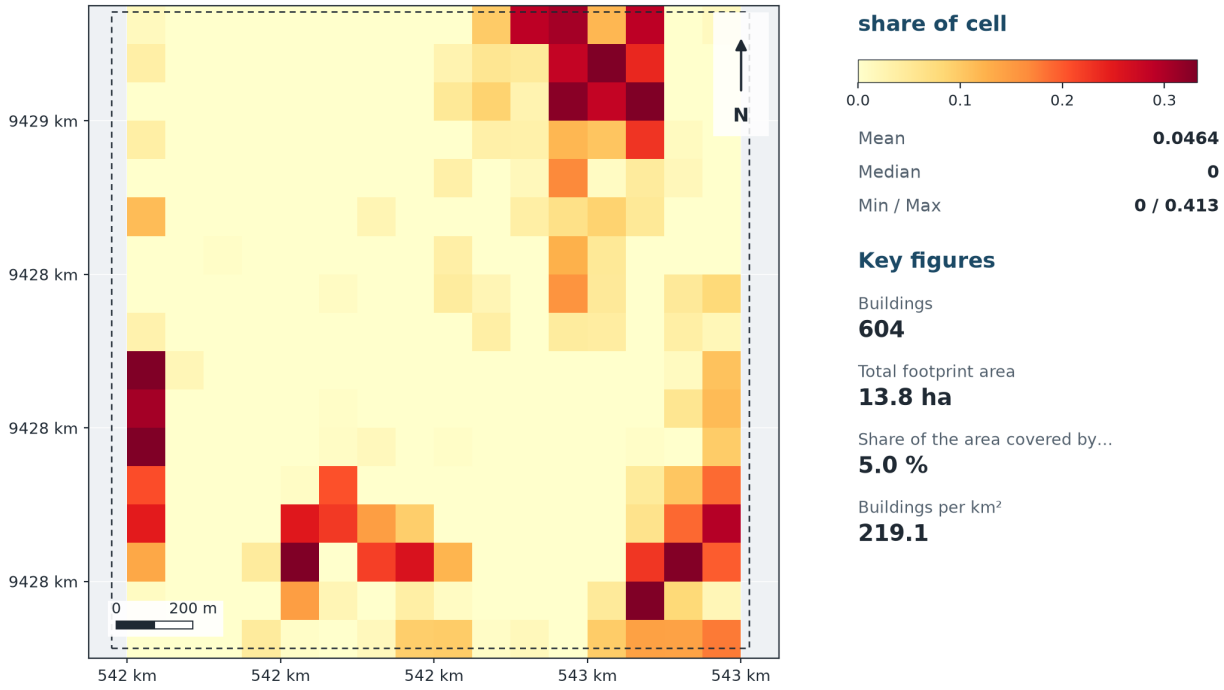
Study area and inputs

Area of interest (west, south, east, north)	-80.6250, -5.1800, -80.6100, -5.1650
Centre	-5.1725°, -80.6175°
Area of interest	2.76 km²
Period	n/a (published dataset)
Output grid	18 × 17 pixels, EPSG:32717

Results

Building Footprints

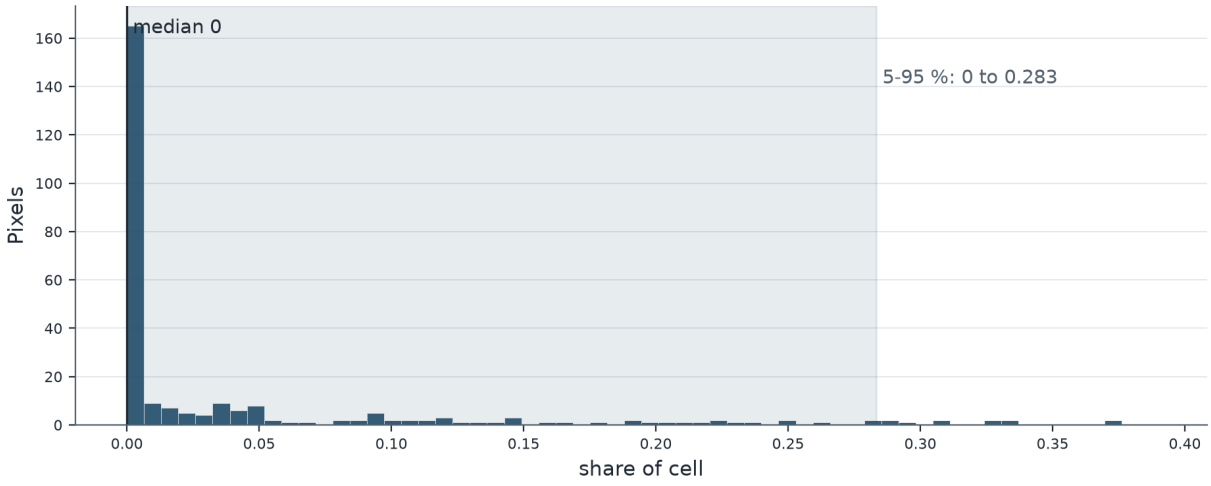
Share of each cell covered by buildings · n/a (published dataset) · AOI -80.6250, -5.1800, -80.6100, -5.1650



CRS: EPSG:32717 · Areas measured from the output raster · No basemap · PyGeoVision 2.2.0 · 2026-10-30 07:12 UTC

Map: Building Footprints

Distribution: Share of each cell covered by buildings



Summary chart

Key statistics (reported by the pipeline)

Measure	Value
Buildings	604
Total footprint area	13.8 ha
Share of the area covered by buildings	5.0 %
Buildings per km²	219.1
Mean footprint (m²)	228.9
Median footprint (m²)	141.0
Highest cell coverage	41.3 %
Area (km²)	2.76

Other statistics

Measure	Value
Source	microsoft
Cell m	100
Footprints file	piura_out/footprints_microsoft/building_footprints.gpkg
Buildings with height	0
Mean height m	n/a

Value statistics: Share of each cell covered by buildings

Statistic	Value
Minimum (share of cell)	0.000
5th percentile (share of cell)	0.000
25th percentile (share of cell)	0.000
Median (share of cell)	0.000
Mean (share of cell)	0.046
75th percentile (share of cell)	0.045
95th percentile (share of cell)	0.283
Maximum (share of cell)	0.413
Standard deviation (share of cell)	0.089

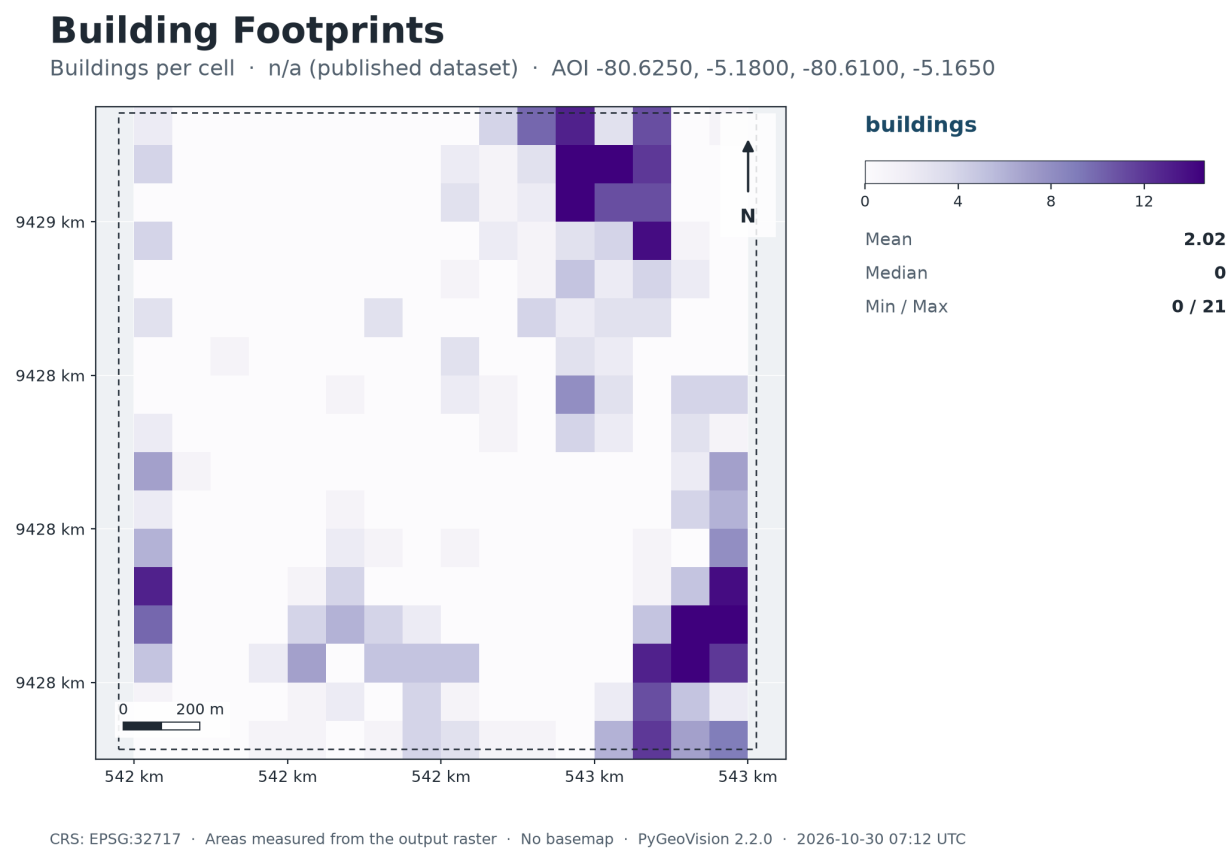
Over 272 valid pixels.

Value statistics: Buildings per cell

Statistic	Value
Minimum (buildings)	0.000
5th percentile (buildings)	0.000
25th percentile (buildings)	0.000
Median (buildings)	0.000
Mean (buildings)	2.022
75th percentile (buildings)	2.000
95th percentile (buildings)	11.450
Maximum (buildings)	21.000
Standard deviation (buildings)	3.773

Over 272 valid pixels.

Additional layers



Map: Buildings per cell

Method and parameters

Where are the buildings in the area, how many, and how much ground do they cover?

Published footprints (default): the polygons of Microsoft's Global ML Building Footprints, Google's Open Buildings v3 or OpenStreetMap are read for the box, clipped to it and measured in the local UTM zone. Nothing is detected on your imagery.

building_density.tif is the share of each cell covered by building area (a building counts in the cell holding its centroid); building_count.tif counts buildings per cell.

With a checkpoint the pipeline instead runs a segmentation network you trained (class 1 = building) and writes building_mask.tif.

Pipeline	building_footprints
Area of interest	-80.6250, -5.1800, -80.6100, -5.1650
cell_m	100
source	microsoft
period	n/a (published dataset)

Interpretation and limitations

How to read the result

- Dataset footprints differ in date: a recent building may be missing and a demolished one still present.

Limitations

- Published footprints are only as complete, accurate and current as their source; compare the three sources if it matters.
- A building_mask.tif result (only when you pass a checkpoint) is the output of a network you trained and has not been checked against reference data; at 10 m individual houses are not resolvable.

Deliverables

Files

File	Type	Contents
piura_out/footprints_microsoft/building_density.tif	GeoTIFF raster	Share of each cell covered by buildings
piura_out/footprints_microsoft/building_count.tif	GeoTIFF raster	Buildings per cell
piura_out/footprints_microsoft/deliverables/building_footprints_report.pdf	PDF report	Complete analysis report
piura_out/footprints_microsoft/deliverables/building_footprints_report.xlsx	Excel workbook	Complete analysis report
piura_out/footprints_microsoft/deliverables/building_footprints_report.docx	Word report	Complete analysis report
piura_out/footprints_microsoft/deliverables/maps/building_footprints_map.png	PNG image	Map or chart for slides and documents
piura_out/footprints_microsoft/deliverables/maps/building_footprints_map_count.png	PNG image	Map or chart for slides and documents
piura_out/footprints_microsoft/deliverables/maps/building_footprints_histogram.png	PNG image	Map or chart for slides and documents
piura_out/footprints_microsoft/deliverables/vectors/building_footprints.gpkg	GeoPackage (all layers)	Vector features with attributes
piura_out/footprints_microsoft/deliverables/vectors/geojson/building_footprints_aoi.geojson	GeoJSON, EPSG:4326	Vector features with attributes
piura_out/footprints_microsoft/deliverables/vectors/geojson/building_footprints_buildings.geojson	GeoJSON, EPSG:4326	Vector features with attributes
run_summary.json	JSON	Machine-readable record of this run

Vector layers

Layer	Features	Total area (ha)	Note
buildings	604	13.8	

Vector layers made from rasters carry class_id, class_name, color_hex, area_ha, area_m2 (equal-area projection), pct_aoi, cx, cy, pipeline and period; layers a pipeline wrote itself (buildings, roads, streams, fire points) keep their own attributes. Coordinates are WGS 84 (EPSG:4326). The GeoPackage also holds the 'aoi' outline and, for raster-derived layers, a dissolved '<layer>_classes' summary.

References

- Microsoft. Global ML Building Footprints (github.com/microsoft/GlobalMLBuildingFootprints), Open Data Commons Open Database License.
- Sirko, W. et al. (2021). Continental-scale building detection from high resolution satellite imagery. arXiv:2107.12283. (Google Open Buildings v3.)
- OpenStreetMap contributors. Data (c) OpenStreetMap, available under the Open Database License (openstreetmap.org/copyright).